



STEEL TAKE-OFF & ROD SCHEDULE

Fe550 TMT (ARS) · 12 m stock rods · Preliminary — for ordering

Project **ID-1369** · Client **Mr. Murali / Mr. Sathyamoorthi**

Drawings: DEEPSpace 1369-01 → 1369-13, Rev 01 dtd 2026-09-07

Structural Engineer: Er. Nishanth · Prepared 2026-09-08

STRUCTURE SUMMARY (FROM DRAWINGS)

Structure	G + 1 (RCC framed, upto terrace)	Concrete / Steel	M20 · Fe550
Built-up area	GF 1,014 + FF 1,014 = 2,028 sq ft	Lap length	57 × dia (tension)
Columns	15 nos (C1×2, C2×3, C3×6, C4×4)	Clear cover	25 beam / 40 column / 50 footing (mm)
Footings	14 nos (F1×2, F2×2, F3×6, F4×3, CF1×1)	UG Sump	4'6" × 10'0" × 8'0" = 10,000 L

A. ROD REQUIREMENT — ORDER SUMMARY

8 MM 272 rods · 1,289 kg	10 MM 197 rods · 1,459 kg	12 MM 110 rods · 1,173 kg	16 MM 81 rods · 1,536 kg
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TOTAL — ARS Fe550 TMT, 12 m rods (incl. 3% cutting waste)

660 rods · 5.46 T

DIA	NET LENGTH	RODS (NET)	RODS +3%	WEIGHT	PRINCIPAL USE
8 mm	3,161 m	264	272	1,289 kg	Column ties, beam stirrups, slab distribution
10 mm	2,287 m	191	197	1,459 kg	Footing mesh, slab main + extra top, sump
12 mm	1,278 m	107	110	1,173 kg	Beam main bars, column bars, curtail bars
16 mm	933 m	78	81	1,536 kg	Column main bars, main beam bars
TOTAL	7,659 m	640	660	5,458 kg	≈ 5.46 tonnes

B. ELEMENT-WISE / FLOOR-WISE BREAK-UP

#	ELEMENT	8 MM	10 MM	12 MM	16 MM	WEIGHT	% OF TOTAL
1	Columns — 15 nos, footing → terrace	594 m	—	359 m	533 m	1,395 kg	26.5%
2	Footings — 14 nos incl. CF1	—	472 m	—	—	291 kg	5.5%
3	Plinth beams — PB1/PB2/PB3	370 m	—	399 m	—	501 kg	9.5%
4	GF roof beams — GFB1–5, CB1	487 m	6 m	239 m	210 m	740 kg	14.0%
5	FF roof beams — FFB1–4, CB1	474 m	6 m	273 m	190 m	735 kg	14.0%
6	Slabs — GF + FF roof (2 × 1,014 sf)	1,236 m	1,338 m	—	—	1,314 kg	24.9%
7	UG Sump — 10,000 L	—	465 m	8 m	—	294 kg	5.6%
NET TOTAL (before wastage)		3,161 m	2,287 m	1,278 m	933 m	5,271 kg	100%

C. COLUMN-WISE DETAIL

MARK	SIZE	MAIN BARS (EACH)	TIES	NOS	12 MM	16 MM	8 MM TIES	WEIGHT
C1 stub	9" × 9"	4 – Y12	Y8 @ 8" c/c	2	23 m	—	21 m	29 kg
C2	9" × 1'0"	4 – Y16 + 2 – Y12	Y8 @ 8" c/c	3	53 m	107 m	117 m	262 kg
C3	9" × 1'3"	4 – Y16 + 4 – Y12	Y8 @ 8" c/c	6	212 m	213 m	273 m	633 kg
C4	9" × 1'3"	6 – Y16 + 2 – Y12	Y8 @ 8" c/c	4	71 m	213 m	182 m	472 kg
TOTAL COLUMNS				15	359 m	533 m	594 m	1,395 kg

D. FOOTING-WISE DETAIL — ALL Y10

TYPE	L × B × D	BOTTOM MESH	COLUMN	NOS	10 MM LENGTH	WEIGHT
F1	3'6" × 3'6" × 1'0"	Y10 @ 8" c/c b/w	0'9" × 0'9"	2	28 m	17 kg
F2	4'6" × 4'0" × 1'3"	Y10 @ 7" c/c b/w	1'0" × 0'9"	2	53 m	32 kg
F3	4'9" × 4'0" × 1'6"	Y10 @ 6" c/c b/w	1'3" × 0'9"	6	198 m	122 kg
F4	5'0" × 4'0" × 1'6"	Y10 @ 5" c/c b/w	1'3" × 0'9"	3	124 m	76 kg
CF1	7'3" × 5'0" × 1'6"	Y10 @ 5" c/c b/w	2 columns	1	70 m	43 kg
TOTAL FOOTINGS				14	472 m	291 kg

E. UG SUMP — 4'6" × 10'0" × 8'0" (10,000 L)

ITEM	REINFORCEMENT	LENGTH	WEIGHT
Wall — horizontal	Y10 @ 6" c/c	150 m	93 kg
Wall — vertical	Y10 @ 6" c/c	179 m	111 kg
Base raft	Y10 @ 6" c/c both ways	55 m	34 kg
Cover slab	Y10 @ 4" c/c both ways	81 m	50 kg
Extra around manhole (2'×2')	2 – Y12 top & bottom	8 m	7 kg
TOTAL UG SUMP		473 m	294 kg

⚠ **NOT INCLUDED — data missing from the drawing set. Add before placing the final order.**

- **Septic tank** — sheet 1369-04 is an un-filled template ("L×B×H = 0000 Ltrs"). Reinforcement spec is given (walls Y10@6", cover slab Y10@4" b/w) but **no dimensions**. Send me the size and I will add it — expect roughly **250–350 kg** for a typical 2-chamber tank.
- **Lintels & sunshades** — sections LB1–LB4 are given on sheet 1369-05, but there is no door/window schedule to count them against. Typically **200–300 kg** for this size of house.
- **Staircase** — no stair reinforcement sheet in this set. Typically **150–250 kg** per flight.
- **Second floor (551 sq ft)** — listed on sheets 1369-03/05 but no beam/slab drawings issued.
- **Compound wall, parapet, water tank** — not in this structural set.

Allowing for these, budget ≈ **6.2 – 6.5 tonnes** overall against the 5.46 T computed here.

Assumptions used (please confirm before ordering):

- **Storey heights are not shown anywhere in the drawing set.** Assumed: footing top → plinth 8'6"; plinth → GF roof 10'0"; GF roof → FF roof 10'0" (total column ≈ 28'6"). Stub column C1 = 9'0". **Every 1 ft change in floor height moves column steel by ≈ 45 kg.**
- Built-up area taken as **1,014 sq ft per floor** (title block on sheets 02/04/06/08/09/11/12). Sheets 03 and 05 carry a different set of areas (1,709 / 1,599 / 551 = 3,859 sf) — **please confirm which is correct.**
- Beam runs scaled off the grid (1–2 = 4'11", 2–3 = 11'0", 3–4 = 11'0"; E–D = 7'9", D–B = 18'1½", B–A = 12'9").
- Slab steel from legend: main Y10 @ 7" c/c, distribution Y8 @ 6" c/c, extra top Y10 @ 8" c/c over 30% of area.
- Laps at 57 d per drawing note 5; one lap per 12 m stock joint. Hooks 12 d, stirrup hooks 10 d.
- Cutting wastage 3%. Rod length 12 m (ARS standard). Unit weight d²/162 kg/m.

Cross-check: 5,271 kg over 2,028 sq ft = **2.60 kg/sq ft**, which sits in the normal band for an economical G+1 framed structure with 9" columns (2.5 – 4.0 kg/sq ft). No red flag in the quantities.